

Control mode: velocity-mode OFFBOARD throughout (runtime drive *and* two-phase entry). `rpp` → `twist_to_setpoint` → `setpoint_raw/local` (type_mask VELOCITY 3527) → PX4 . **No position controller in the live path** — `OFFBOARD_POSITION_MODE_PLAN.md` is a deferred plan, not what runs.

Completed & field-proven	verified in-tree	Pending — genuinely on main's lineage	not here
Placement	GPS-surveyed → live EKF (P+L-R_anchor) · <i>mission_placement.py:131</i>	Point-nav	No point-mission stack (Task_01); DXF/polyline only
Entry	Two-phase velocity entry, spray-OFF, 2-point · <i>offboard_controller.py:373</i>	GPS CSV	No lat/lon point-mission ingest (QGC-waypoint→polyline only)
Pre-align	Run-0 entry pivot (gated) · <i>rpp_controller_node.py:338</i>	Plan-stage	Point-mission plan-and-stage E2E (depends on above)
Completion	DONE gated on measured coast-down · <i>rpp:1517</i> <i>_hold_at_completion</i>	Spray modes	Only continuous; no dash / point mode
Scale gate	Ref double-scale guard · <i>path_engine/engine.py:422</i>	Spray API	No per-path mode sidecar / hot-apply to live controller
Mission clear	POST /api/mission/clear (ABORTED/ERROR) · <i>routes/mission.py:137</i>	Spray ident	No path-fingerprint identity binding / strict gating
Auth	Password login + scoped tokens + disable bypass · <i>auth.py:133</i>	Spray gate	No GPS_SURVEYED runtime spray gate (see Attention)
Spray safety	Actuator state machine, latency, startup OFF · <i>spray_controller_node.py:803</i>	Param contract	Continuous-mode <code>gps_*/obstacle_*/point_*</code> hardening (N/A — params absent)
RTK	NTRIP/LoRa self-heal + watchdog age-gate · <i>rtk_manager.py:224</i>	Mission ops	Skip / restart / operation coordinator / terminal cleanup / event journal
Recorder	SHA256 bundle + manifest + retention · <i>bag_autorecord.py:255</i>	Joystick V2	No joystick_controller / manual-control socket event
Telemetry	Read-only monitoring API · <i>routes/telemetry.py:20</i>	MANUAL_CONTROL	float32 path + rejection logging
Partial / differs from main		Throttle caps	Joystick steering/throttle tuning caps
Densify	MARK 5 cm but TRANSIT/connectors 15 cm; no global ≤5 cm pass · <i>straight_line.py:52</i>	Bridge yaw-hold	Corner-stop / reverse-brake yaw hold for joystick
Fail-closed	Baseline watchdog + D3 latch present; mainline spray-backend pack absent	Swagger	X-Rover-Token OpenAPI security scheme
GPS 8-dp	Non-gap: float64 passthrough exceeds 8 dp, no explicit round	Activity CSV	/activity export (JSON only today)
Needs attention — promote first		AsyncAPI	Socket.IO contract docs
SAFETY	Spray gate has no GPS_SURVEYED provenance check — degraded-GPS mission can still spray · <i>spray_controller_node.py:775</i>	GCS matrix	Migration compatibility matrix
Footgun	entry_prealign_enabled resets to False every rpp-pipeline restart (P4)	measured_speed	Telemetry measured_speed_m_s field
P1	Arc doesn't flow (~68% stop-pivot); check 5/15 cm densify split, no knob retune	3D label	3D GPS fix label + plaintext-auth rejection
		Capture race	Bag e-stop→503 retry fix + GPS nuisance e-stop debounce
		Do NOT rebuild (frozen by invariant)	
		#3	Densified 5 cm entry — velocity 2-point entry is the design
		#40	Corner-stop param retunes — keep 0.50 / 0.08 / no-hold frozen (12/14)
		Pos-mode	Whole position-mode entry — demoted; velocity is primary